Week 3 Update

Michael Fang

March 2026

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Key new results this week:

- New deterministic substitution tiling with $\alpha = 2$ (first of its kind)
- Algebraic proof that $2 < \alpha < 3$ is inaccessible with 3×3 matrices
- Generalized metrics C_{II}, A_{III} for Classes II and III
- Stealthy patterns have two distinct phases with very different $\bar{\Lambda}$
- $\bar{\Lambda} = 1/6$ for integer lattice is exact (no finite-N correction)

Recap

Pattern	Cl.	α	$\bar{\Lambda}$
Integer Lattice	I	∞	0.167
Fibonacci	I	3.0	0.201
Silver	I	3.0	0.250
Bronze	I	3.0	0.282
URL ($a = 1$)	I	2.0	0.333
Stealthy ($\chi = 0.3$)	I	∞	0.357
Bombieri-Taylor	I	1.545	0.377
Period-Doubling	II	1.0	div.
0222	III	0.64	div.

Open questions entering this week:

- Can the $2 < \alpha < 3$ gap be filled by a quasicrystal?
- Bombieri-Taylor: spreadability fit gives $\hat{\alpha} \approx 1.9$, not 1.545 — why?
- No metric analogous to $\bar{\Lambda}$ exists for Classes II and III
- How do estimates converge with N?

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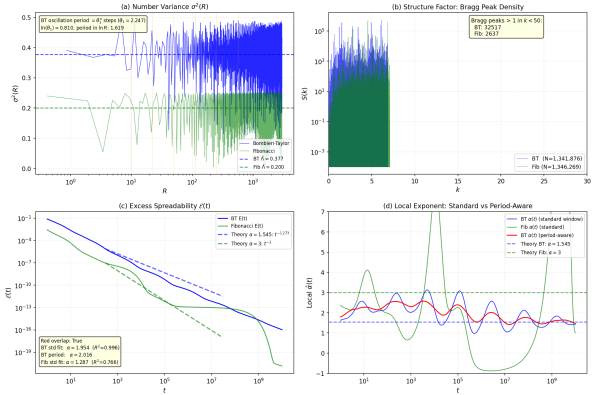
BT eigenvalue $\alpha = 1.545$ is exact from the formula $1 - 2\ln|\lambda_2|/\ln\lambda_1$; the spreadability fit was unreliable. All 2×2 metallic-mean matrices satisfy $|\det M| = 1$, forcing $|\lambda_2| = 1/|\lambda_1|$ and $\alpha = 3$. Stealthy: $\alpha \to \infty$ because $S(k) = 0$ over a finite interval, not a power law.

Why Spreadability Fails for Bombieri-Taylor

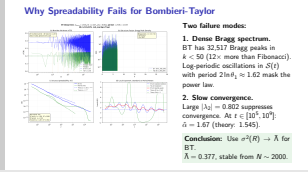


Two failure modes:

1. Dense Bragg spectrum.
BT has 32,517 Bragg peaks in $k < 50$ (12× more than Fibonacci). Log-periodic oscillations in $\mathcal{S}(t)$ with period $2\ln\theta_1 \approx 1.62$ mask the power law.

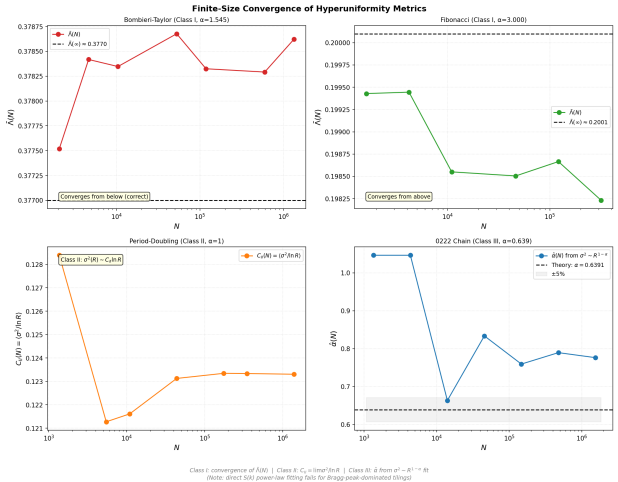
2. Slow convergence.
Large $|\lambda_2| = 0.802$ suppresses convergence. At $t \in [10^5, 10^9]$: $\hat{\alpha} = 1.67$ (theory: 1.545).

Conclusion: Use $\sigma^2(R) \to \bar{\Lambda}$ for BT.
$\bar{\Lambda} = 0.377$, stable from $N \sim 2000$.



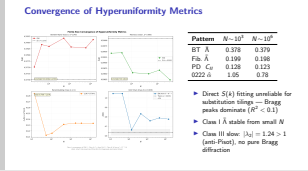
Rod overlap: minimum BT spacing $= 1.000$; need $2a < 1.000$; corrected to $a_{\text{rod}} = 0.45$. Fibonacci has $|\lambda_2| = 0.445 = 1/\theta_1$, much smaller, so its spreadability converges well. The variance-based $\bar{\Lambda} = 0.377$ is reliable and stable.

Convergence of Hyperuniformity Metrics



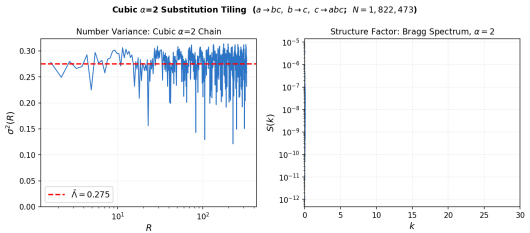
Pattern	$N \sim 10^3$	$N \sim 10^6$
BT $\bar{\Lambda}$	0.378	0.379
Fib. $\bar{\Lambda}$	0.199	0.198
PD C_{II}	0.128	0.123
0222 $\hat{\alpha}$	1.05	0.78

- Direct $S(k)$ fitting unreliable for substitution tilings — Bragg peaks dominate ($R^2 < 0.1$)
- Class I $\bar{\Lambda}$ stable from small N
- Class III slow: $|\lambda_2| = 1.24 > 1$ (anti-Pisot), no pure Bragg diffraction



$S(k)$ for a substitution tiling is a superposition of delta functions; no smooth k^α background. 0222 chain: $|\lambda_2| = \sqrt{5} - 1 > 1$, so diffraction has a diffuse component; $\hat{\alpha}$ converges slowly to theory 0.639. Class I $\bar{\Lambda}$ is a spatial average (stable because large-R fluctuations cancel quickly).

A Deterministic Substitution Tiling with $\alpha = 2$



Substitution: $a \to bc,\ b \to c,\ c \to abc$

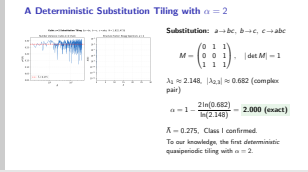
$$M = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad |\det M| = 1$$

$\lambda_1 \approx 2.148,\ |\lambda_{2,3}| \approx 0.682$ (complex pair)

$$\alpha = 1 - \frac{2\ln(0.682)}{\ln(2.148)} = \mathbf{2.000\ (exact)}$$

$\bar{\Lambda} = 0.275$, Class I confirmed.

To our knowledge, the first *deterministic* quasiperiodic tiling with $\alpha = 2$.



Characteristic polynomial: $x^3 - x^2 - 2x - 1 = 0$. $\bar{\Lambda} = 0.275$ lies between Silver (0.250) and Bronze (0.282). Pattern generated to $N \approx 1.8 \times 10^6$ points. The $\alpha = 2$ value is exact by the algebraic argument on the next slide.

The $2 < \alpha < 3$ Gap: An Algebraic Obstruction

Systematic search: All 3×3 non-negative integer matrices with row sums ≤ 4.

39,304 candidates $\rightarrow$ 4,882 Pisot $\rightarrow$ 305 with eigenvalue formula in $(2, 3)$

Every one of the 305 gives $\alpha = 2.000$ or $\alpha = 3.000$ exactly. None fill the gap.

Why? For a 3×3 matrix with $|\det M| = 1$ and a complex conjugate eigenvalue pair $(\lambda_2, \bar{\lambda}_2)$:

$$\lambda_1 |\lambda_2|^2 = |\det M| = 1 \;\Rightarrow\; |\lambda_2| = \frac{1}{\sqrt{\lambda_1}} \;\Rightarrow\; \alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln \lambda_1} = 1 - \frac{\ln(1/\lambda_1)}{\ln \lambda_1} = \boxed{2}$$

All 305 candidates fall into this case. For all-real eigenvalues, the exhaustive search finds no case with $2 < \alpha < 3$.

Conclusion: Filling $2 < \alpha < 3$ requires algebraic degree ≥ 4 (i.e., 4×4 substitution matrices).

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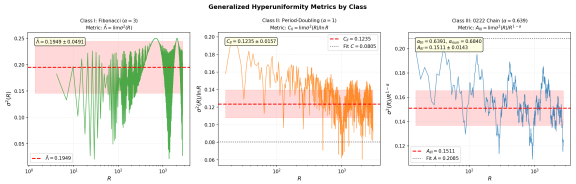
Conclusion: Filling $2 < \alpha < 3$ requires algebraic degree ≥ 4 (i.e., 4×4 substitution matrices).

All primitive 3×3 substitution matrices satisfy $|\det M| = 1$ (Pisot-Vijayaraghavan property). The complex-conjugate case: $\det M = \lambda_1 \cdot \lambda_2 \cdot \bar{\lambda}_2 = \lambda_1 |\lambda_2|^2 = 1$, so $|\lambda_2| = 1/\sqrt{\lambda_1}$ exactly, giving $\alpha = 2$ regardless of the specific value of λ_1. The all-real case can in principle yield $\alpha \neq 2, 3$, but within row sums ≤ 4 no such case exists. A 4×4 search is the natural follow-up.

Generalized Hyperuniformity Metrics for Classes II and III



Normalized variances plateau at large R, confirming asymptotic scaling.

$\bar{\Lambda}$ is finite only for Class I.

Class II $(\sigma^2 \sim C_{II} \ln R)$:

$$C_{II} = \lim_{R\to\infty} \frac{\sigma^2(R)}{\ln R}$$

Period-Doubling:

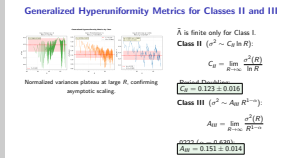
$$\boxed{C_{II} = 0.123 \pm 0.016}$$

Class III $(\sigma^2 \sim A_{III}\, R^{1-\alpha})$:

$$A_{III} = \lim_{R\to\infty} \frac{\sigma^2(R)}{R^{1-\alpha}}$$

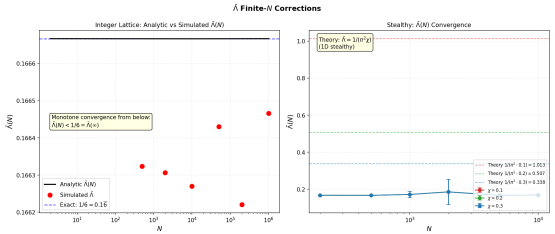
0222 ($\alpha = 0.639$):

$$\boxed{A_{III} = 0.151 \pm 0.014}$$



Period-Doubling: $|\lambda_2| = 1$ exactly ($\alpha = 1$, Class II borderline Pisot). 0222: $|\lambda_2| = \sqrt{5} - 1 > 1$ (not Pisot), hence diffuse component in $S(k)$. C_{II} requires $N > 10^5$ to converge; A_{III} convergence is slow due to $|\lambda_2| > 1$. Open: does the $\bar{\Lambda}$–C_{II} correspondence remain continuous as $\alpha \to 1^+$?

Finite-N Properties of $\bar{\Lambda}$



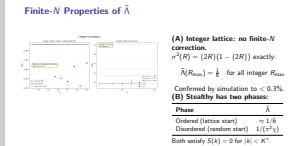
(A) Integer lattice: no finite-N correction.
$\sigma^2(R) = \{2R\}(1 - \{2R\})$ exactly:

$$\bar{\Lambda}(R_{\max}) = \frac{1}{6} \quad \text{for all integer } R_{\max}$$

Confirmed by simulation to $< 0.3\%$.
(B) Stealthy has two phases:

Phase	$\bar{\Lambda}$
Ordered (lattice start)	$\approx 1/6$
Disordered (random start)	$1/(\pi^2\chi)$

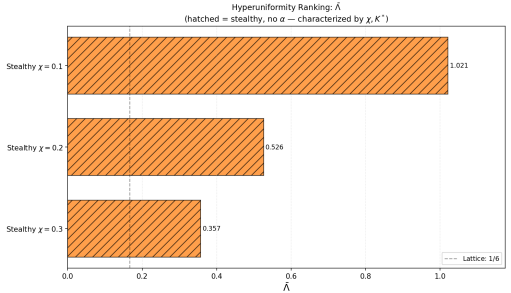
Both satisfy $S(k) = 0$ for $|k| < K^*$.



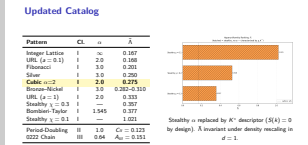
Lattice: $\sigma^2(R) = \{2R\}(1 - \{2R\})$ integrates to $1/12$ over each half-unit cell $[n/2, (n+1)/2]$, so the running average is exactly $1/6$ for any integer $R_{\max}$. The two stealthy phases: starting from a perturbed lattice, the collective-coordinate optimizer finds the near-lattice ground state ($\bar{\Lambda} \approx 1/6$); starting from random positions (as in the collaborator data) it finds the disordered phase ($\bar{\Lambda} = 1/(\pi^2\chi)$). Both are valid stealthy hyperuniform configurations satisfying the same $S(k) = 0$ constraint.

Updated Catalog

Pattern	Cl.	α	$\bar{\Lambda}$
Integer Lattice	I	∞	0.167
URL ($a = 0.1$)	I	2.0	0.168
Fibonacci	I	3.0	0.201
Silver	I	3.0	0.250
Cubic α=2	**I**	**2.0**	**0.275**
Bronze–Nickel	I	3.0	0.282–0.310
URL ($a = 1$)	I	2.0	0.333
Stealthy $\chi = 0.3$	I	—	0.357
Bombieri-Taylor	I	1.545	0.377
Stealthy $\chi = 0.1$	I	—	1.021
Period-Doubling	II	1.0	$C_{II} = 0.123$
0222 Chain	III	0.64	$A_{III} = 0.151$



Stealthy α replaced by K^* descriptor ($S(k) = 0$ by design). $\bar{\Lambda}$ invariant under density rescaling in $d = 1$.



Density invariance: rescaling $x_i \to x_i/\rho$ leaves $\bar{\Lambda}$ unchanged (verified numerically, diff < 0.001). Metallic means: Fib 0.201, Ag 0.250, Br 0.282, Cu 0.293, Ni 0.310 — monotone, converging. The URL cloaked value $\bar{\Lambda} = 1/3$ is a natural upper bound for the metallic-mean sequence. Stealthy $K^* = 2\pi\rho\chi$.

Summary and Open Questions

New results:

- Bombieri-Taylor: spreadability unreliable (log-periodic oscillations + slow convergence); $\bar{\Lambda} = 0.377$ is the reliable metric
- Direct $S(k)$ power-law fitting fails for substitution tilings; variance-based metrics are stable from small N
- New deterministic substitution tiling with $\alpha = 2$, $\bar{\Lambda} = 0.275$ (first of its kind)
- $2 < \alpha < 3$ is algebraically inaccessible with 3×3 unimodular matrices; requires degree ≥ 4
- Generalized metrics: $C_{II} = 0.123$ (Period-Doubling), $A_{III} = 0.151$ (0222 chain)
- Integer lattice: $\bar{\Lambda} = 1/6$ exactly at all N; stealthy has ordered ($\bar{\Lambda} \approx 1/6$) and disordered ($\bar{\Lambda} = 1/(\pi^2\chi)$) phases

Open questions:

- Can 4×4 substitution matrices fill $2 < \alpha < 3$?
- Is $\bar{\Lambda} = 1/4$ exact for the Silver chain?
- What is the exact limiting value of the metallic-mean $\bar{\Lambda}$ sequence?

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The algebraic gap result is the central theoretical contribution: unimodular 3×3 matrices with a complex-conjugate pair are pinned to $\alpha = 2$ exactly by the determinant constraint. Silver $\bar{\Lambda} = 1/4$ exact would parallel lattice (1/6) and possibly Fibonacci (≈ 0.201). Metallic-mean limit: URL cloaked value 1/3 is a natural candidate.

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